Circle the correct answer for each question. If you're unsure, you can write numbers 1, 2, 3, etc. next to the choices to get partial credit.

Question permutation table:

Version		1	2	3	4
Question					
1	→	4	1	4	1
2*	→	5	2	5	2
3	→	1	3	1	3
4*	→	2	4	2	4
5*	→	3	5	3	5

*: Question follows previous

The following table shows the responses from a sample of 680 people in the General Social Survey to the question, "Do you sometimes drink more than you think you should?"

Gender	Yes	No	Total
Male	151	177	328
Female	92	260	352
Total	243	437	680

1. What proportion of males responded that they drink more than they should?

   (a) 151/243
   (b) 151/328
   (c) 243/680
   (d) 151/680

   Correct answers: (b)

   Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(d)	(c)	(a)	(d)
(b)	→	(c)	(b)	(c)	(a)
(c)	→	(a)	(d)	(d)	(b)
(d)	→	(b)	(a)	(b)	(c)

2. Does there appear to be an association between gender and whether a person responds that they drink more than they should?

   (a) Yes, since a different number of people of each gender are in the sample.
   (b) Yes, since the percent that responded that they drink more than they should differs from 50
   (c) Yes, since the proportion who responded they drink more than they should differs between males and females.
   (d) No, there does not appear to be an association between these two variables.

   Correct answers: (c)

   Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(a)	(d)	(c)	(d)
(b)	→	(d)	(c)	(b)	(a)
(c)	→	(c)	(a)	(a)	(b)
(d)	→	(b)	(b)	(d)	(c)

A study published in 2007 by Christopher Johnson, professor of music education and music therapy at the University of Kansas, revealed that students in elementary schools who took private music lessons scored around 20 percent higher in math scores on standardized tests, compared to students who did not take private music lessons. Use this to answer the following three questions:

3. What is the explanatory variable?

   (a) Music therapy
   (b) Whether a student took private music lessons
   (c) Math score on standardized test
   (d) Grade level

   Correct answers: (b)

   Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(a)	(b)	(a)	(d)
(b)	→	(c)	(d)	(b)	(a)
(c)	→	(d)	(a)	(d)	(c)
(d)	→	(b)	(c)	(c)	(b)

4. What is the response variable?

   (a) Music therapy
   (b) Whether a student took private music lessons
   (c) Math score on standardized test
   (d) Grade level

   Correct answers: (c)

   Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(b)	(b)	(b)	(a)
(b)	→	(d)	(c)	(d)	(d)
(c)	→	(a)	(d)	(a)	(b)
(d)	→	(c)	(a)	(c)	(c)

5. Which of the following could be a potential confounding variable?

   (a) Household income
   (b) Socio-economic status
   (c) Whether the parents hired a math tutor
   (d) All of these

   Correct answers: (d)

   Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(a)	(d)	(b)	(d)
(b)	→	(b)	(b)	(c)	(a)
(c)	→	(c)	(a)	(a)	(b)
(d)	→	(d)	(c)	(d)	(c)